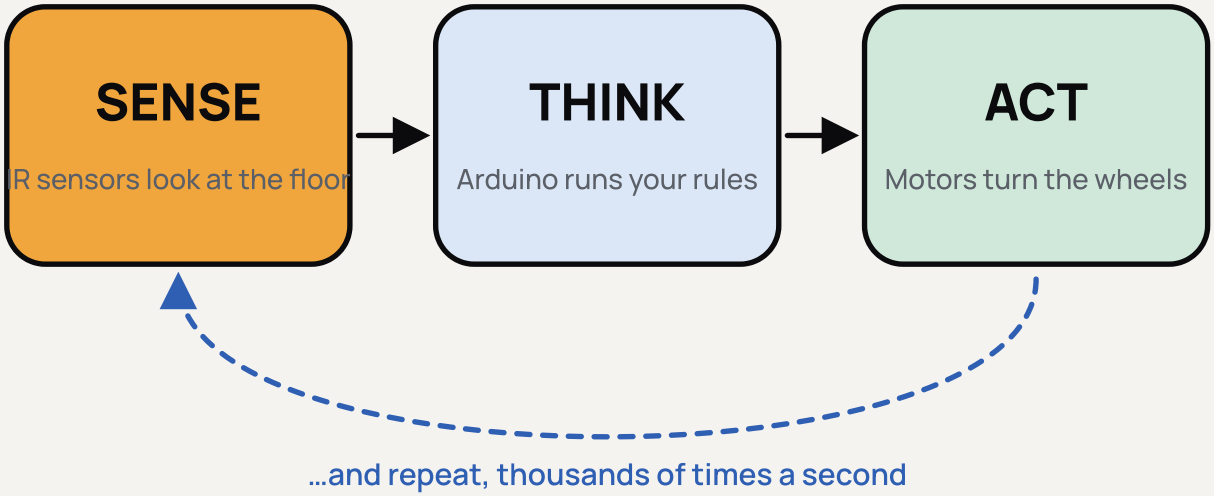
Component identification chart

Every part in the kit, its name and its job

Part	Job	Good to know
Arduino Uno	The brain: runs your program and makes decisions	Keep it on spacers; never touch the pins with power on
USB cable (A–B)	Uploads programs from the computer	Unplug before running on batteries
IR sensor module ×2	The eyes: see black or white below	Blue knob sets sensitivity; LED shows what it sees
L298N motor driver	The muscles' helper: powers the motors	Gets warm when working: that is normal
DC gear motor ×2	Turn the wheels	Left motor = motor A, right motor = motor B
Wheels ×2 and castor	Carry the robot	The castor keeps it balanced at the front
Battery pack (2 × 18650, 7.4 V)	Energy for motors and Arduino	Adult supervision; never short the wires
Switch	Turns everything on and off	Keep it OFF while wiring
Chassis and screws	The body	Keep screws in a bowl
Jumper wires	Carry power and signals	Red = power, black = ground by habit

How the parts work together

Every robot repeats three steps



Wiring: battery → driver → motors, Arduino → driver

